



Server and Instance Types

- One goal of capacity planning is to make growth and shrinkage of capacity predictable. Greatly improve your chances by standardizing on a few hardware types and then well characterizing those platforms.
- In a cloud infrastructure such as Amazon Web Server, the use of different machine instance sizes is an attempt to create standard servers.
- Reducing the variability between servers makes it easier to troubleshoot problems and simpler to deploy and configure new systems.
- As much as possible, you also should assign servers standardized roles and populate those servers with identical services.
- A server with the same set of software, system configuration, and hardware should perform similarly if given the same role in an infrastructure.



Server and Instance Types

- There is more performance variability in cloud machine instances than you might expect. This variability may be due to your machine instances or storage buckets being moved from one system to another system.
- Therefore, you should be attentive to the increase in potential for virtual system variability and build additional safety factors into your infrastructure. System instances in a cloud can fail, and it is up to the client to recognize these failures and react accordingly.
- Virtual machines in a cloud are something of a black box, and you should treat them as such because you have no idea what the underlying physical reality of the resources you are using represents.
- Capacity planning seeks to compare the capability of one system against another system and to choose the solution that is not only the right size but provides the service with the best operational parameters at the lowest cost.
- In Figure a graph of different server types is shown, with some hypothetical physical servers plotted against Amazon Machine Instance types.
- This type of graph allows you to add the appropriate server type to your infrastructure while performing a cost analysis of the deployment.



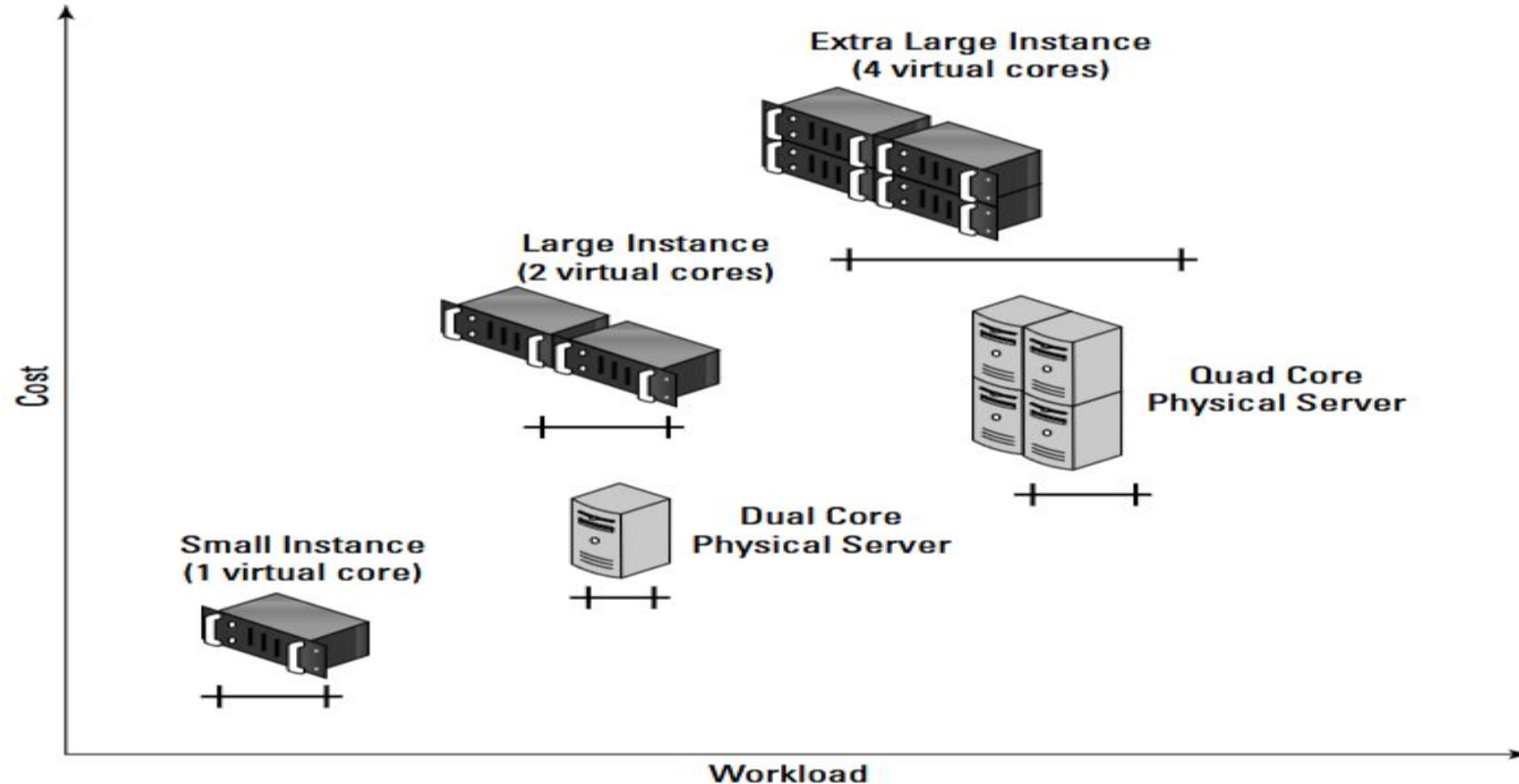
An Amazon Machine Instance (AMI) is described as follows:

- **Micro Instance:** 633 MB memory, 1 to 2 EC2 Compute Units (1 virtual core, using 2 Cus for short periodic bursts) with either a 32-bit or 64-bit platform
- **Small Instance (Default):** 1.7GB memory, 1 EC2 Compute Unit (1 virtual core with 1 EC2 Compute Unit), 160GB instance storage (150GB plus 10GB root partition), 32-bit platform, I/O Performance: Moderate, and API name: m1.small
- **High-Memory Quadruple Extra Large Instance:** 68.4GB of memory, 26 EC2 Compute Units (8 virtual cores with 3.25 EC2 Compute Units each), 1,690GB of instance storage, 64-bit platform, I/O Performance: High, and API name: m2.4xlarge
- **High-CPU Extra Large Instance:** 7GB of memory, 20 EC2 Compute Units (8 virtual cores with 2.5 EC2 Compute Units each), 1,690GB of instance storage, 64-bit platform,
- **I/O Performance:** High, API name: c1.xlarge

Server and Instance Types

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Relative costs and efficiencies of different physical and virtual servers





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- While you may not know what exactly EC2 Compute Unit or I/O Performance High means, at least we can measure these performance claims and attach real numbers to them.
- Amazon says that an EC2 Compute Unit is the equivalent of a 1.0-1.2 GHz 2007 Opteron or 2007 Xeon processor, but adds that “Over time, we may add or substitute measures that go into the definition of an EC2 Compute Unit, if we find metrics that will give you a clearer picture of compute capacity.
- What they are saying essentially is that this was the standard processor their fleet used as its baseline, and that over time more powerful systems may be swapped in.
- Whatever the reality of this situation, from the standpoint of capacity planning, all you should care about is measuring the current performance of systems and right-sizing your provisioning to suit your needs.
- In cloud computing, you can often increase capacity on demand quickly and efficiently.
- Not all cloud computing infrastructures automatically manage your capacity for you as Amazon Web Service’s Auto Scaling (<http://aws.amazon.com/autoscaling/>) feature does.



Server and Instance Types

- For example, Amazon Web Services' infrastructure runs not only clients' machine instances, but also Amazon.com's machine instances as well as Amazon.com's partners' instances.
- If you need more virtual servers on Black Monday to service Web sales, you may be out of luck unless you have an SLA that guarantees you the additional capacity.
- Finding an SLA that you can rely on is still something of a crap shoot. If you are in the position of having to maintain the integrity of your data for legal reasons, this imposes additional problems and constraints.
- Easy to get started and requires little up-front costs. However, cloud computing is not cheap. At the point where you have a large cloud-based infrastructure, you may find that it is more expensive to have a metered per-use cost than to own your own systems.
- The long-term efficiencies are realized in the reduced need for IT staff and management, the ability to react quickly to business opportunities, and the freeing of a business from managing networked computer systems to concentrate on their core businesses.
- The Total Cost of Ownership (TCO) of a cloud-based infrastructure might be beneficial, it might be difficult to convince the people paying the bills that this is really so.



Any cloud-computing system resource is difficult to plan for, it is network capacity. There are three aspects to assessing network capacity:

- Network traffic to and from the network interface at the server, be it a physical or virtual interface or server
- Network traffic from the cloud to the network interface
- Network traffic from the cloud through your ISP to your local network interface (your computer)
- This makes analysis complicated. You can measure factor 1, the network I/O at the server's interface with system utilities, as you would any other server resource.
- For a cloud-based virtual computer, the network interface may be a highly variable resource as the cloud vendor moves virtual systems around on physical systems or reconfigures its network pathways on the fly to accommodate demand.
- To measure network traffic at a server's network interface, you need to employ what is commonly known as a network monitor, which is a form of packet analyzer.
- Microsoft includes a utility called the Microsoft Network Monitor as part of its server utilities,



The site Sectools.org has a list of packet sniffers at <http://sectools.org/sniffers.html>.

- Wireshark (<http://www.wireshark.org/>), formerly called Ethereal
- Kismet (<http://www.kismetwireless.net/>), a WiFi sniffer
- TCPdump (<http://www.tcpdump.org/>)
- Dsniff (<http://www.monkey.org/~dugsong/dsniff/>)
- Ntop (<http://www.ntop.org/>)
- EtherApe (<http://etherape.sourceforge.net/>)



Factor 2 is the cloud's network performance, which is a measurement of WAN traffic. A WAN's capacity is a function of many factors:

- Overall system traffic (competing services)
- Routing and switching protocols
- Traffic types (transfer protocols)
- Network interconnect technologies (wiring)
- The amount of bandwidth that the cloud vendor purchased from an Internet backbone provider Again, factor 2 is highly variable and unlike factor 1, it isn't easy to measure in a reliable way.
- Tools are available that can monitor a cloud network's performance at geographical different points and over different third-party ISP connections.
- This is done by establishing measurement systems at various well-chosen network hubs.



Network Capacity

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- A company that makes WAN network monitoring software, has set up a series of these points of presence at various Internet hubs and uses its networking monitoring software called PathView Cloud to collect data in a display that it calls the Cloud Performance Scorecard (<http://www.apparentnetworks.com/CPC/scorecard.aspx>).
- Figure shows this Web page populated with statistics for some of the cloud vendors that Apparent Networks monitors.
- Use PathView Cloud as a hosted service to evaluate your own cloud application's network performance at these various points of presence and to create your own scorecard of a cloud network.
- Current pricing for the service is \$5 per network path per month. The company also sells a small appliance that you can insert at locations of your choice and with which you can perform your own network monitoring.
- The last factor, factor 3, is the connection from the backbone through your ISP to your local system, a.k.a.
- Internet connection is more like an intelligently managed thin straw that you are desperately trying to suck information out of. So Factor 3 is measurable, even if the result of the measurement isn't very encouraging, especially to your wallet.



Network Capacity

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- Internet connectivity over the last mile (to the home) is the Achilles heel of cloud computing. The scarcity of high-speed broadband connections (particularly in the United States) and high pricing are major impediments to the growth of cloud computing.
- Many organizations and communities will wait on the sidelines before embracing cloud computing until faster broadband becomes available.
- Indeed, this may be the final barrier to cloud computing's dominance. That's one of the reasons that large cloud providers like Google are interested in building their own infrastructure, in promoting high-speed broadband connectivity, and in demonstrating the potential of high-speed WANs.
- Google is running a demonstration project called Google Fibre for Communities (<http://www.google.com/appserve/fiberrfi/public/overview>) that will deliver 1 gigabit-per-second fiber to the home.
- The few lucky municipalities chosen (with 50,000 to 500,000 residents) for the demonstration project will get advanced broadband applications, many of which will be cloud-based.



Apparent Networks' Cloud Performance Center provides data on WAN throughput and uptime at Internet network hubs.





Scaling

- In capacity planning, after you have made the decision that you need more resources, you are faced with the fundamental choice of scaling your systems.
- You can either scale vertically (scale up) or scale horizontally (scale out), and each method is broadly suitable for different types of applications.
- To scale vertically, you add resources to a system to make it more powerful. For example, during scaling up, you might replace a node in a cloud-based system that has a dual-processor machine instance equivalence with a quad-processor machine instance equivalence.
- You also can scale up when you add more memory, more network throughput, and other resources to a single node. Scaling out indefinitely eventually leads you to an architecture with a single powerful supercomputer.
- Vertical scaling allows you to use a virtual system to run more virtual machines (operating system instance), run more daemons on the same machine instance, or take advantage of more RAM (memory) and faster compute times.
- Applications that benefit from being scaled up vertically include those applications that are processor-limited such as rendering or memory-limited such as certain database operations—queries against an in-memory index.



Scaling

- **Horizontal scaling** or scale out adds capacity to a system by adding more individual nodes.
- In a system where you have a dual-processor machine instance, you would scale out by adding more dual-processor machines instances or some other type of commodity system.
- Scaling out indefinitely leads you to an architecture with a large number of servers (a server farm), which is the model that many cloud and grid computer networks use.
- Horizontal scaling allows you to run distributed applications more efficiently and is effective in using hardware more efficiently because it is both easier to pool resources and to partition them.
- Although your intuition might lead you to believe otherwise, the world's most powerful computers are currently built using clusters of computers aggregated using high speed interconnect technologies such as InfiniBand or Myrinet.
- Scale out is most effective when you have an I/O resource ceiling and you can eliminate the communications bottleneck by adding more channels. Web server connections are a classic example of this situation.



Scaling

- These broad generalizations between scale up and scale out are useful from a conceptual stand point, but the reality is that there are always tradeoffs between choosing one method for scaling your cloud computing system versus the other.
- The pricing model that cloud computing vendors now offer their clients isn't fully mature at the moment, and you may find yourself paying much more for a high-memory extra-large machine instance than you might pay for the equivalent amount of processing power purchased with smaller system equivalents.
- This has always been true when you purchase physical servers, and it is still true when purchasing virtual servers. Cost is one factor to pay particular attention to, but there are other tradeoffs as well.
- Scale out increases the number of systems you must manage, increases the amount of communication between systems that is going on, and introduces additional latency to your system.